# SCE 672: Advanced Robotics

## Term Project Assessment 2: Differential Flatness Pursuit and Motor Voltage Dynamics

**Student:** Ameen Ur Rehman (202527430) **Marks:** 10
**Instructor:** Prof. Aamer Iqbal Bhatti
**Institution:** King Fahd University of Petroleum & Minerals
**Code and data:** `https://github.com/AmeenUrRehman/RoboMaster-Advanced-Robotics` (folder `Assignment 2`)
**Demonstration video 1:** `https://youtu.be/EaZKQfd3usE`
**Demonstration video 2:** `https://youtu.be/JlHnuVnLLrA`

---

**Abstract**

A RoboMaster EP searches for a floor target (Tag 3), aligns with it, follows a smooth trajectory generated with differential flatness, stops 20 cm in front of it, and logs the four wheel-motor voltages once per second. The solution is a ROS 2 package (`rm_flatness_nav`) built from small, tested pieces: a closed-form flatness inversion and a cubic boundary-value solver, a four-state supervisory state machine, an AprilTag detector with pose recovery, a wheel-rate and armature-voltage model, and a ROS 2 node that connects them to the robot. All hardware results come from real logs on the robot; every terminal figure is rendered from the captured text of the real command, and every plot comes from the real CSV files. Three complete autonomous runs over USB ended with the state `ARRIVED`, each with all four voltages exactly 0.000 V at the halt and a peak speed of about 0.137 m/s (limit 0.16 m/s). Run 5, the one used for Table 1 and the plot, used the three printed tags lying flat on the floor as the assignment describes; runs 3 and 4 used the tag standing upright. The stop gap, measured by hand, was 20 cm in runs 3 and 4 (to the upright tag); in run 5 it was about 20 cm to the edge of the A4 sheet, which is about 30 cm to the black square and about 38 cm to the tag centre (see Section 4), i.e. the robot stopped about 18 cm short of the planned 20 cm to the tag centre. The report also documents what did *not* work (two earlier runs that never halted and several flat-tag attempts with a wrong tag scale), what was learned from the camera (it is fixed to the chassis and does not tilt with the arm), and where the setup deviates from the assignment text (an upright tag in runs 3 and 4, a start distance of 1.2 to 1.5 m instead of 1.5 to 1.8 m, and a stop that comes from the planned trajectory rather than from a camera reading).

### Summary of results

| Quantity | Result |
|---|---|
| Target | Tag 3, `tag36h11` ID 3, black square 0.15 m; run 5 lying flat (Tags 1, 2 beside it), runs 3–4 upright |
| Trajectory horizon (Eq. 3) | run 5: $T = 1.6\,(1.05 - 0.20)/0.16 = 8.5$ s planned, about 8.9 s measured (1 Hz log); run 4: 10.8 s planned, 10.9 s measured |
| Peak forward speed (runs 3 to 5) | 0.137 m/s (0.136 in run 5, 1 Hz samples); limit 0.16 m/s |
| Final motor voltages | $V_1 = V_2 = V_3 = V_4 = 0.000$ V at the halt |
| Measured stop gap (tape measure) | 20 cm in runs 3 and 4; run 5: about 20 cm to the sheet edge (about 30 cm to the black square) |
| Search / align / chase duration | run 5: 8.0 s / 2.0 s / 8.9 s (total 18.9 s); run 4: 11.4 s / 2.5 s / 10.9 s (25.2 s) |
| Unit tests | 18 of 18 pass |
| Camera tilt (measured from images) | about 20° downward, fixed to the chassis |

## Mission and system overview

### The mission

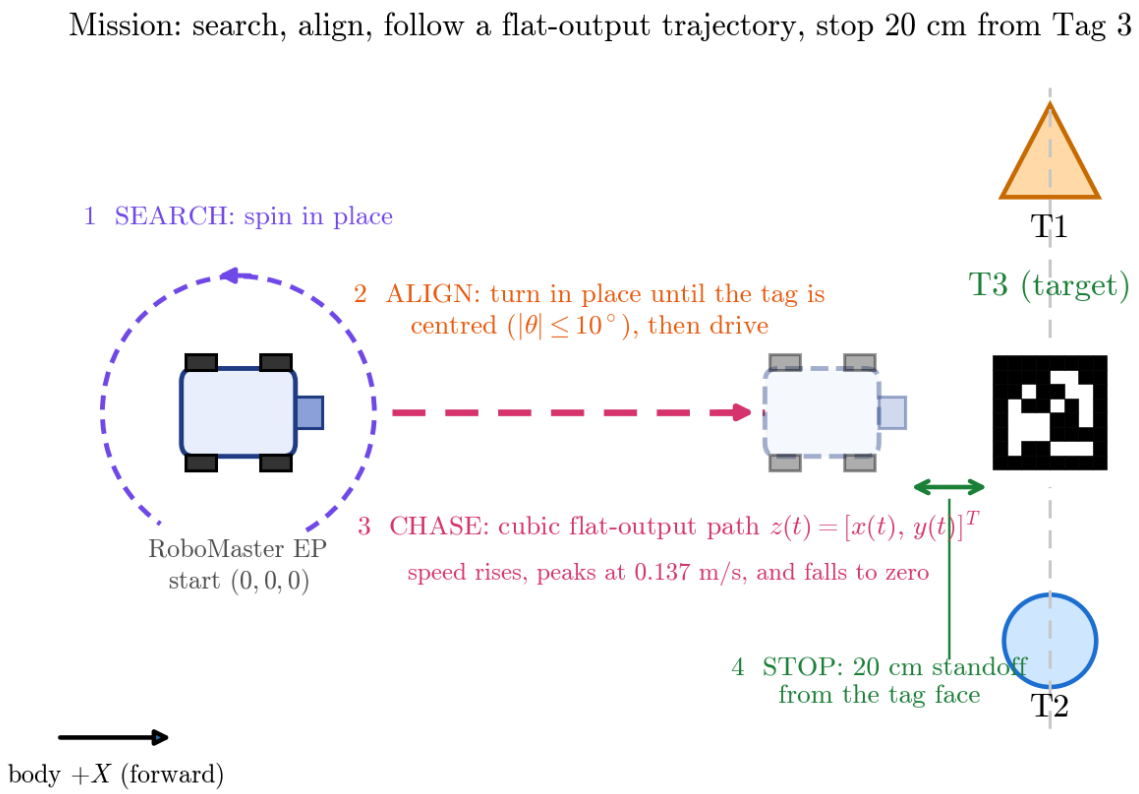



**Figure 1:** The four behaviours. The tags are drawn as simple shapes for clarity (Tag 1 triangle, Tag 2 circle, Tag 3 the square AprilTag); in the experiments only Tag 3, a real `tag36h11` AprilTag, was placed in front of the robot. The flat-output path of phase 3 is a straight line in the robot frame because the robot first turns until the tag is centred.

### The tags

The assignment names three targets, $T_1$, $T_2$ and $T_3$, without specifying a marker family or size. The detector therefore uses the standard AprilTag family `tag36h11`, which OpenCV provides, with the identifier as a parameter (`tag_id`, default 3). Printable A4 sheets for the three tags (IDs 1, 2, 3, black square exactly 150 mm) were generated with `Assets/make_assets.py`; each PDF was rendered back and decoded with OpenCV to check the identifier and the 150 mm size (measured 150.1 mm). The experiments used Tag 3 only.

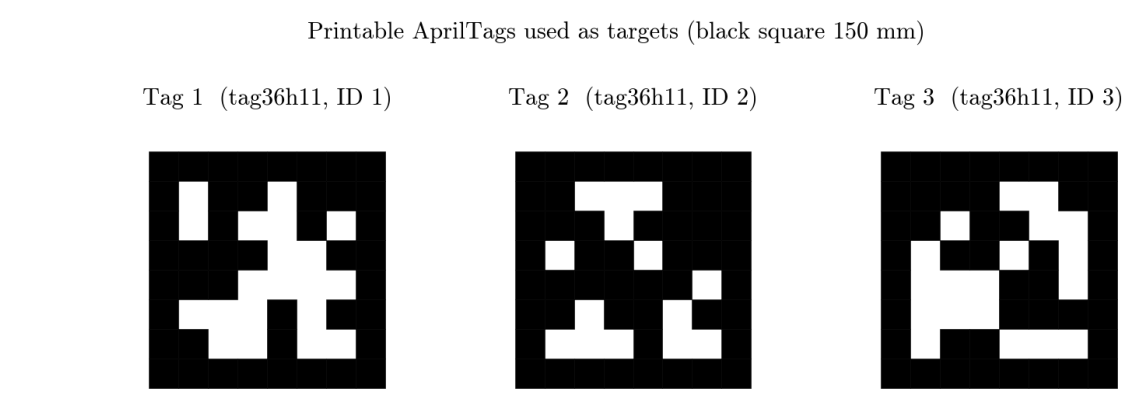



**Figure 2:** Printable AprilTags (`tag36h11`, IDs 1, 2, 3).

### Software architecture

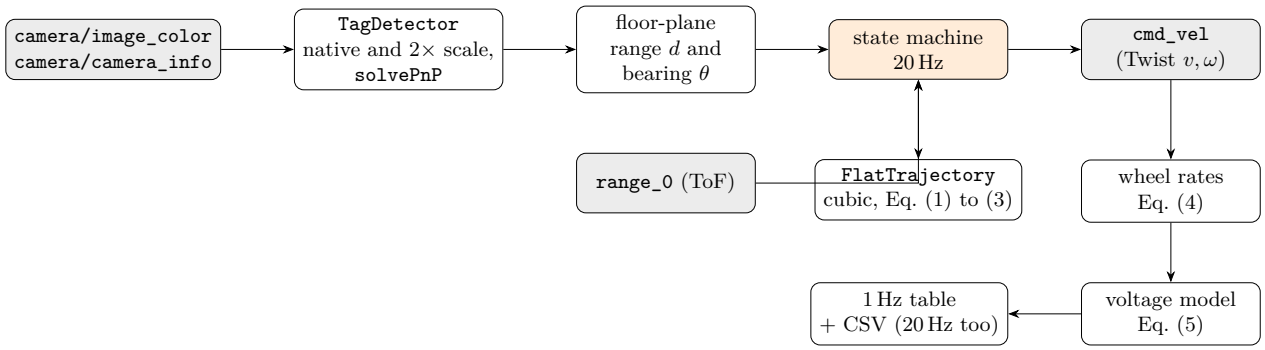



**Figure 3:** Data flow of `flatness_nav`. Grey boxes are ROS interfaces of the `robomaster_ros` driver.

| File | Role |
|---|---|
| `flatness.py` | flatness inversion, `solve_flat_polynomial` (`numpy.linalg.solve`), time horizon, `FlatTrajectory` |
| `wheel_voltage.py` | wheel rates (Eq. 4) and the armature voltage model (Eq. 5) |
| `state_machine.py` | SEARCHING, ALIGNING, CHASING, ARRIVED (pure logic, unit tested) |
| `tag_detector.py` | AprilTag detection, pose with `solvePnP`, floor-plane range and bearing |
| `nav_node.py` | the ROS 2 node: camera, ToF, `cmd_vel`, logging |
| `camera_pose.py`, `sim.py` | visibility predictor; offline simulation (not hardware data) |

The control loop runs at 20 Hz. The code is in the repository (see the link at the top) and the main files are listed in Appendix A.

### Data provenance and run history

Four real runs of the full system exist. They are listed because the two failures led directly to design changes. The columns are computed from the logs by `scripts/make_tables.py`.

| Run | logged | link | final state | search (s) | first seen (m) | chase $d_0$ (m) | chase (s) | peak $v$ (m/s) | last seen (m) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 16:00 | Wi-Fi AP | SEARCHING (no halt) | 2.3 | 0.906 | 0.906 | – | 0.137 | 0.535 |
| 2 | 16:04 | Wi-Fi AP | SEARCHING (no halt) | 44.9 | 1.189 | 1.061 | – | 0.137 | 0.605 |
| 3 | 18:36 | USB | ARRIVED | 35.9 | 1.375 | 1.367 | 11.8 | 0.137 | 0.379 |
| 4 | 19:14 | USB | ARRIVED | 11.4 | 1.280 | 1.280 | 10.9 | 0.137 | 0.351 |
| 5 | 00:44 | USB | ARRIVED | 8.0 | 1.067 | 1.053 | 8.9 | 0.136 | 0.465 |

**Run history.** The four real runs. "First seen" and "last seen" are the camera distances at the first and last detection that appear in the 1 Hz log.

- **Runs 1 and 2 (before the fix).** The robot detected the tag, aligned and started the chase, but when the tag left the image it returned to SEARCHING and never halted. In those runs the helper output of the session shows a configured tag size of 0.20 m while the tag is 0.15 m, so the distances reported there are about $0.20/0.15 = 1.33$ times too large.
- **The fix.** If the tag disappears *during* a chase and was last seen within 1.8 m, the planned trajectory simply continues from its time base (it already ends at the standoff).
- **Runs 3, 4 and 5 (after the fix, over USB, tag size 0.15 m).** All three ended in `ARRIVED`. Runs 3 and 4 used the tag standing upright (20 Hz logs); run 5 used the printed tags lying flat (only the 1 Hz log was kept) and is the one shown in Table 1 and the main plot.

## Problem 1: Visual search, in-place centering and standoff guard

### (i) Downward camera setup

**Why the camera must look down.** The tags lie on the floor, so the useful part of the field of view is the floor. A level camera at height $h$ sees a floor point at distance $d$ at a depression angle $\arctan(h/d)$. With the vertical half field of view of the stream (about 30°, from $f_y \approx 316$ px at 360 p), a level camera loses the floor in front of it once $\arctan(h/d) > 30°$, i.e. closer than $h/\tan 30°$, so the tag would leave the image before the 20 cm standoff. Tilting the camera down by $\beta$ moves the visible floor band closer, to $h/\tan(30° + \beta)$, and makes the view of the floor less grazing, which reduces the foreshortening of a flat tag.

**What was measured.** The driver was commanded to the pose $(X, Y) = (105, 30)$ mm (the driver names the vertical axis $z$) and the arm read back $x = 0.105$ m, $z = 0.03$ m. Camera frames were then captured at this pose and at other arm positions (Figure 4): the frames *did not change* when only the arm moved. The robot description in TF attaches the camera to the arm, but on this robot the camera is fixed to the chassis, so its tilt is not set by the arm pose, and that part of TF is not used. In the images the horizon (the bases of the far walls) lies roughly 100 to 130 px above the image centre, which corresponds to a downward tilt of about $20° \pm 3°$ (the tilt is read from the images and is an estimate). The gripper claw is visible at the bottom centre of every frame, so a floor tag directly ahead and closer than roughly 0.3 m is hidden by it. The arm position $(105, 30)$ mm is a low, retracted pose; the ToF beam clearance was checked from the run log: the sensor module was enabled and in the whole run it never returned a reading below its maximum range, so neither the arm nor the gripper was in its beam. The camera height above the floor was not measured.

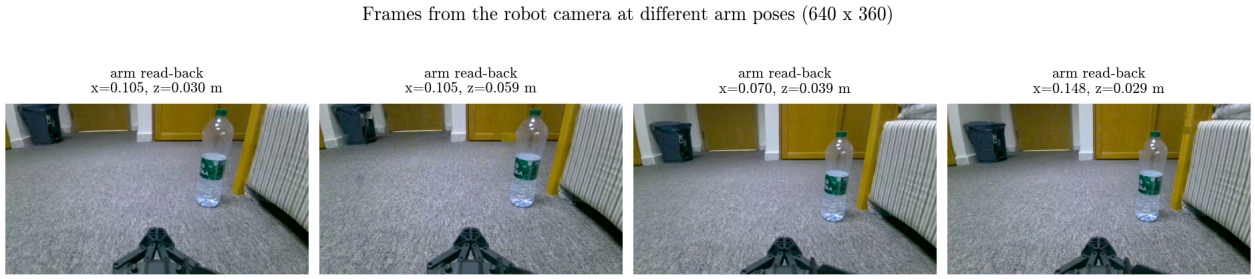



**Figure 4:** Real frames at four arm poses (the arm read-back is printed above each frame). Within each pair of frames (left two, right two) the arm moved and the image did not change. The two pairs differ slightly from each other (the horizon is higher in the left pair), possibly because the robot was moved between the captures.

**The detector on a real frame.** Figure 5 shows the project detector decoding the real tag from a frame of this camera. At that frame the tag is 61 px wide and the pose estimate gives a range of 0.81 m.

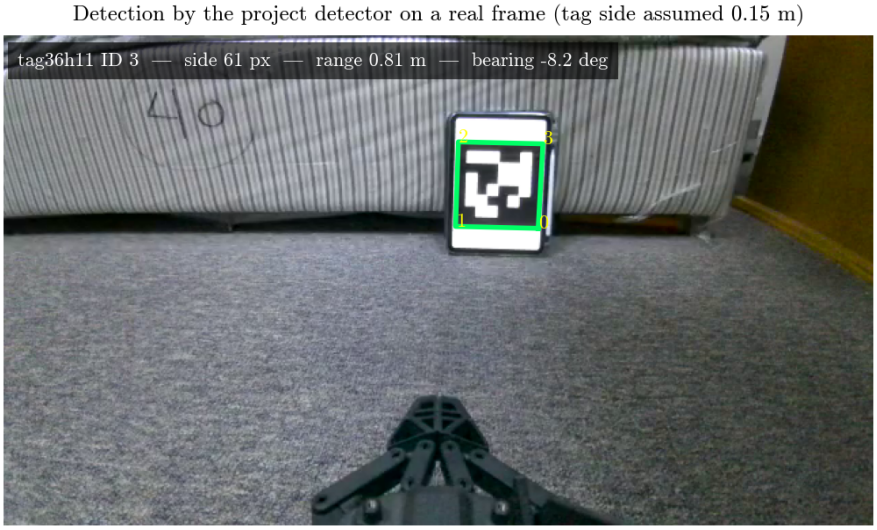



**Figure 5:** Detection of `tag36h11` ID 3 on a real frame. The distance assumes a 0.15 m black square and was not checked with a tape.

**Why the tag was upright, and how far it is visible.** A flat floor tag seen from a camera about 0.25 m above the floor is strongly foreshortened: for a 15 cm tag the model of Figure 6 (camera height assumed) gives only 11 px of height at 1 m and 5 px at 1.5 m, which a detector cannot read. An upright tag keeps its full width: $317 \cdot 0.15/d$ pixels, i.e. 47 px at 1 m. The detection limit was measured by shrinking a real frame: the tag was found at 41 px at native size and not at 33 px; after enlarging the image twice before detection it was found at 33 px and 27 px and not at 24 px. The detector therefore tries native size first and the doubled image if nothing is found (21 ms per empty frame). Later tests with flat tags (Section on flat tags) showed that they are nevertheless readable out to about 0.6 to 1 m, which allowed run 5.

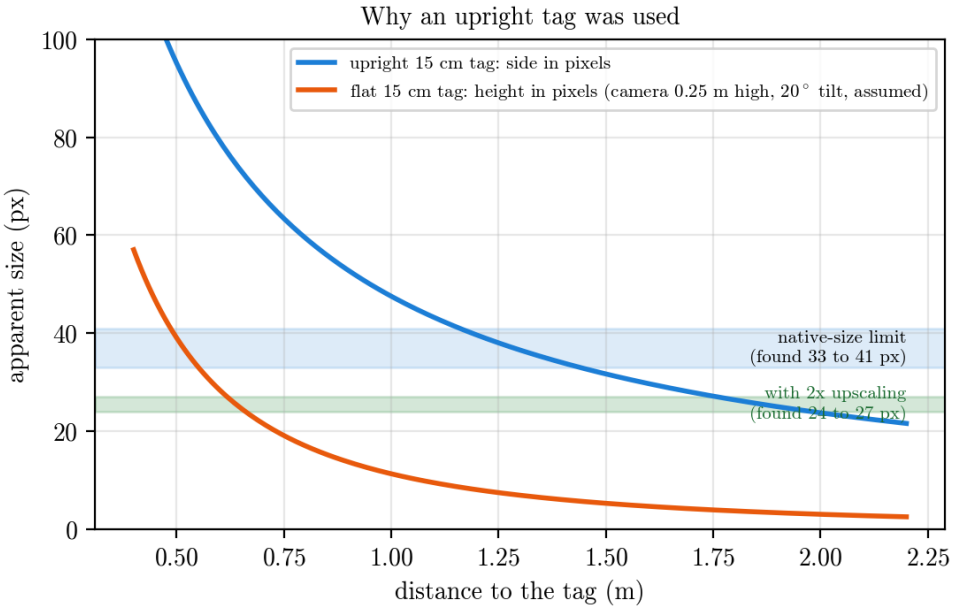



**Figure 6:** Apparent size of a 15 cm tag against distance for an upright and a flat tag, with the measured detection limits. The flat-tag curve uses an assumed camera height of 0.25 m.

### (ii) Search and centering finite state machine

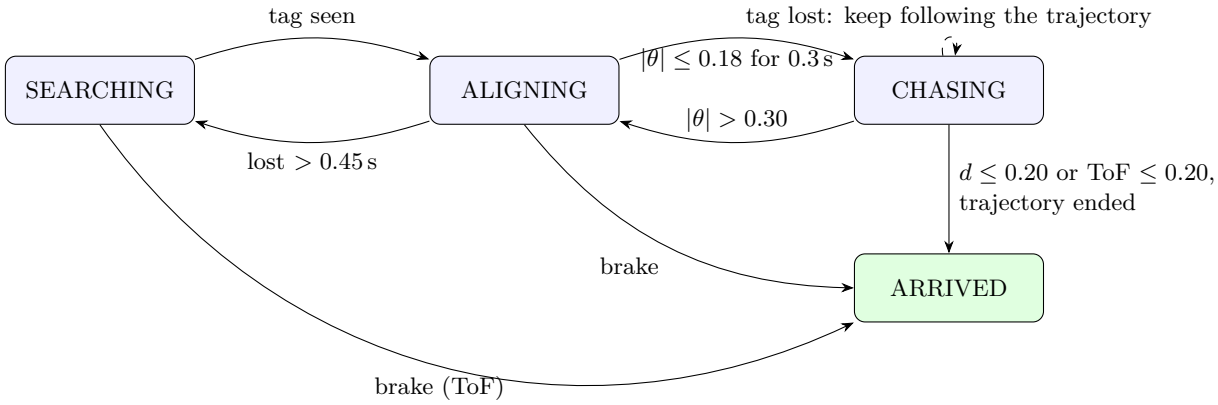



**Figure 7:** State machine of `state_machine.py`. The brake test is made first in every control tick.

| Parameter | Value | Meaning |
|---|---|---|
| `lost_timeout` | 0.45 s | no tag for longer than this: SEARCHING |
| `search_rate` | 0.30 rad/s | in-place rotation, toward the side where the tag was last seen |
| `align_tol` | 0.18 rad (10.3°) | tag centred: start chasing |
| `settle_time` | 0.3 s | bearing must stay inside the tolerance this long |
| `align_gain`, limits | 0.8 $s^{-1}$; 0.12 to 0.40 rad/s | $\omega_z = \mathrm{sat}(K\theta)$ while aligning |
| `realign_tol` | 0.30 rad | hysteresis: leave CHASING only beyond this |
| `chase_bearing_gain` | 0.8 $s^{-1}$ | small correction $K\theta$ added to the flat-output $\omega$ |
| `brake_distance` | 0.20 m | camera range or ToF threshold |
| `blind_chase`, range | true, 1.8 m | continue the trajectory if the tag leaves the image |

**Sign convention.** The bearing $\theta > 0$ means the tag is to the *left* of the camera axis (ROS REP-103, $+\omega_z$ counter-clockwise), so aligning is $\omega_z = +K\theta$. In SEARCHING the direction of the sweep is the side on which the tag was last seen, and counter-clockwise if it has never been seen.

**Why align in place before translating.** The planned path is a straight cubic in the robot frame (the boundary conditions require $y(0) = y(T) = 0$), so it assumes the tag lies straight ahead. If the robot drove while its heading was off by $\theta$, the open-loop path would end laterally displaced by about $d \sin\theta$: for the real run ($d = 1.28\,\text{m}$) a 10° error already gives $1.28 \sin 10° = 22\,\text{cm}$, which is larger than the tag itself. The tag would also drift toward the edge of the image (the half field of view is about 45°), where it is partly clipped or lost. Rotating first removes the heading error at zero translation, so the planned and executed paths coincide and the tag stays near the image centre. Both real runs show this: after the first two seconds of the chase the logged bearing error stayed within 0.02 rad.

### (iii) Standoff braking

The brake is the first test in every control tick, before any other state logic: if the camera ground distance is $d \le 0.20\,\text{m}$ or the front ToF reads $\le 0.20\,\text{m}$, the state becomes ARRIVED and $(v, \omega) = (0, 0)$ is commanded immediately and the state is latched. This behaviour is covered by unit tests (`test_fsm_brake_on_camera_and_tof`). **On the robot neither sensor brake was triggered**: in both runs the upright tag left the top of the image at about 0.35 to 0.38 m, and the ToF never returned a reading below its maximum range, so the stop came from the end of the planned trajectory (state `ARRIVED (trajectory end, tag out of view)`), which is designed to end exactly at the standoff. The measured gap of 20 cm confirms that this worked in these two runs.

**Video stalls.** The runs used the driver's own decoder and the published topic `camera/image_color`. The driver log contains no decoder or buffer errors (its only warning is that the optional FFmpeg transport package is not installed). The optional direct-capture mode of the node (`camera_source:=cv2`) implements the recommended grab thread with a `time.sleep(0.010)` between `cap.grab()` calls; it was not needed and was not used.

## Problem 2: Differential flatness trajectory generation

### (i) Flatness inversion

For a unicycle $\dot{x} = v\cos\theta$, $\dot{y} = v\sin\theta$, $\dot{\theta} = \omega$, the position $z(t) = [x(t), y(t)]^T$ is a flat output: the speed is $v = \sqrt{\dot{x}^2 + \dot{y}^2}$ and the heading is $\theta = \text{atan2}(\dot{y}, \dot{x})$, so

$$v = \sqrt{\dot{x}^2 + \dot{y}^2}, \qquad \omega = \dot{\theta} = \frac{d}{dt}\,\text{atan2}(\dot{y}, \dot{x}) = \frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{\dot{x}^2 + \dot{y}^2}. \tag{1}$$

At standstill ($\dot{x}^2 + \dot{y}^2 \to 0$) the heading is undefined; the implementation returns $\omega = 0$ there. The formula was verified on a circle of radius $R$ traversed at angular rate $\Omega$ (it returns $v = R\Omega$ and $\omega = \Omega$; unit test `test_flat_inversion_circle`).

**(ii) Cubic polynomial matrix solution**

Each component is $z_j(t) = \alpha_0^{[j]} + \alpha_1^{[j]} t + \alpha_2^{[j]} t^2 + \alpha_3^{[j]} t^3$ ($j \in \{x, y\}$) over the horizon $T$. Imposing $z_j(0)$, $\dot{z}_j(0)$, $z_j(T)$, $\dot{z}_j(T)$ gives the linear system

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & T & T^2 & T^3 \\ 0 & 1 & 2T & 3T^2 \end{bmatrix} \begin{bmatrix} \alpha_0^{[j]} \\ \alpha_1^{[j]} \\ \alpha_2^{[j]} \\ \alpha_3^{[j]} \end{bmatrix} = \begin{bmatrix} z_j(0) \\ \dot{z}_j(0) \\ z_j(T) \\ \dot{z}_j(T) \end{bmatrix}, \tag{2}$$

which `solve_flat_polynomial(z0, zdot0, zf, zdotf, T)` solves with `numpy.linalg.solve`. With the boundary values of the assignment, $x(0) = 0$, $\dot{x}(0) = 0.06\,\text{m/s}$, $x(T) = d_{\text{tag}} - 0.20\,\text{m}$, $\dot{x}(T) = 0$ and $y(0) = \dot{y}(0) = y(T) = \dot{y}(T) = 0$, the closed form is

$$\alpha_0 = 0, \quad \alpha_1 = \dot{z}_0, \quad \alpha_2 = \frac{3z_f - 2\dot{z}_0 T}{T^2}, \quad \alpha_3 = \frac{\dot{z}_0 T - 2z_f}{T^3} \qquad (\dot{z}_f = 0),$$

(obtained by solving the last two rows of Eq. (2) for $\alpha_2$ and $\alpha_3$). For $y$ all four boundary values are zero, hence $\alpha^{[y]} = \mathbf{0}$, $y(t) \equiv 0$ and, by Eq. (1), $\omega(t) \equiv 0$: the flat path is a straight line in the robot frame and the speed profile is $v(t) = \dot{x}(t) = \alpha_1 + 2\alpha_2 t + 3\alpha_3 t^2$. The turning during the chase comes only from the small bearing correction.

**(iii) Time horizon scaling**

$$T = \max\!\left(4.0,\ \frac{1.6\,(d_0 - 0.20)}{v_{\max}}\right), \qquad v_{\max} = 0.16\ \text{m/s}. \tag{3}$$

The mean speed of the path is $z_f/T = (d_0 - 0.20)/T = v_{\max}/1.6 = 0.10\,\text{m/s}$, and a cubic that starts slowly and ends at rest peaks at roughly 1.4 times its mean. Because $T$ scales with the distance, the peak is the same for every $d_0$ (the 4 s floor only matters for starts closer than $0.2{+}4{\cdot}0.16/1.6 = 0.6\,\text{m}$). The coefficient table below was computed with the package; the peak is 0.137 m/s in all cases, below 0.16 m/s, and $v(T) = 0$. In the real runs, $d_0 = 1.37\,\text{m}$ (run 3) and 1.28 m (run 4) give $T = 11.7\,\text{s}$ and 10.8 s, and the chases lasted 11.8 s and 10.9 s (the extra 0.1 to 0.2 s is the control-tick granularity).

| $d_0$ (m) | $T$ (s) | $\alpha_2^{[x]}$ (m/s$^2$) | $\alpha_3^{[x]}$ (m/s$^3$) | peak $v$ (m/s) | at $t$ (s) |
|---|---|---|---|---|---|
| 0.90 | 7.00 | 0.02571 | $-28.571 \times 10^{-4}$ | 0.137 | 3.00 |
| 1.00 | 8.00 | 0.02250 | $-21.875 \times 10^{-4}$ | 0.137 | 3.43 |
| 1.28 | 10.80 | 0.01667 | $-12.003 \times 10^{-4}$ | 0.137 | 4.63 |
| 1.50 | 13.00 | 0.01385 | $-8.284 \times 10^{-4}$ | 0.137 | 5.57 |
| 1.80 | 16.00 | 0.01125 | $-5.469 \times 10^{-4}$ | 0.137 | 6.86 |

**Trajectory coefficients.** Cubic coefficients of $x(t)$ and peak speed for several start distances, computed with `FlatTrajectory`.

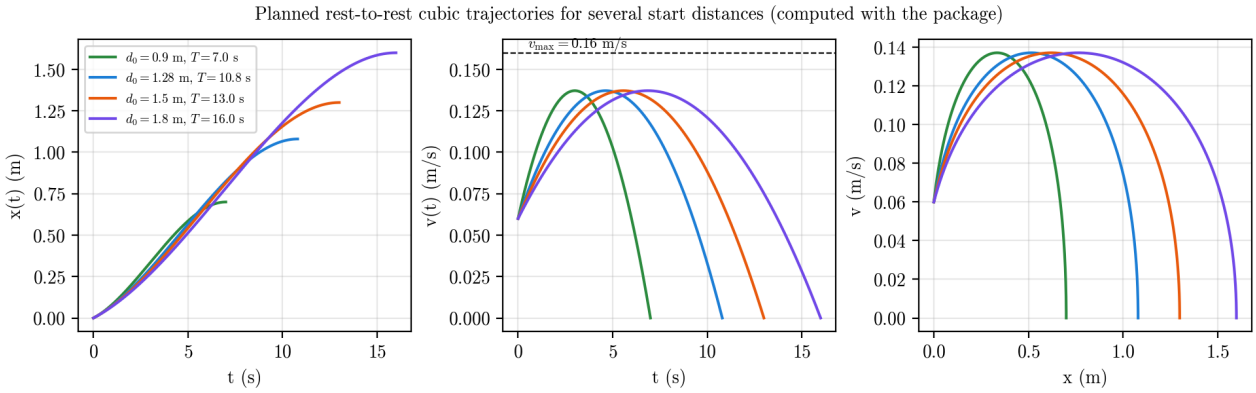



**Figure 8:** Planned $x(t)$, $v(t)$ and $v(x)$ for several start distances.

## Problem 3: Motor voltage telemetry and dynamics logging

### (i) Kinematics to wheel rates

With $r = 0.05\,\text{m}$ and $L_c = 0.20\,\text{m}$, Eq. (4) of the assignment is implemented as

$$\omega_1 = \omega_4 = \frac{v_x + L_c\omega_z}{r}, \qquad \omega_2 = \omega_3 = \frac{v_x - L_c\omega_z}{r} \tag{4}$$

(wheel 1 is front-right, 2 front-left, 3 rear-left, 4 rear-right). Numerical examples that appear in the logs: the in-place search at $\omega_z = 0.30\,\text{rad/s}$ gives $\omega = \pm 1.2\,\text{rad/s}$; the peak cruising speed $v_x = 0.137\,\text{m/s}$ gives $\omega = 2.74\,\text{rad/s}$ ($= 26.2\,\text{rpm}$) on all four wheels.

### (ii) Motor armature voltage model

$$V_i(t) = K_v\,\omega_i(t) + K_a\,\frac{\omega_i(t) - \omega_i(t - \Delta t)}{\Delta t}, \qquad K_v = 0.38\ \text{V s/rad}, \quad K_a = 0.06\ \text{V s}^2\text{/rad}. \tag{5}$$

It is evaluated in every control tick ($\Delta t = 0.05\,\text{s}$) on the wheel rates computed from the *commanded* $(v_x, \omega_z)$. The voltages are therefore model outputs: the motor controller's own voltage is not measured. The first tick uses a zero acceleration term, and once the robot has halted the wheel rates are zero for two consecutive ticks, so $V_i = 0$ exactly (no acceleration term left). Figure 9 splits the logged $V_1$ of run 4 into its two terms: the back-EMF term $K_v\omega_1$ carries the whole cruise (up to 1.05 V), and the inertia term $K_a\Delta\omega_1/\Delta t$ is nonzero only at the velocity steps (starting to turn in SEARCHING, the switch from ALIGNING to CHASING, where the speed jumps to the initial 0.06 m/s, and the final deceleration), reaching about $+1.4\,\text{V}$ at the start of the chase and $-1.3\,\text{V}$ at an in-place reversal.

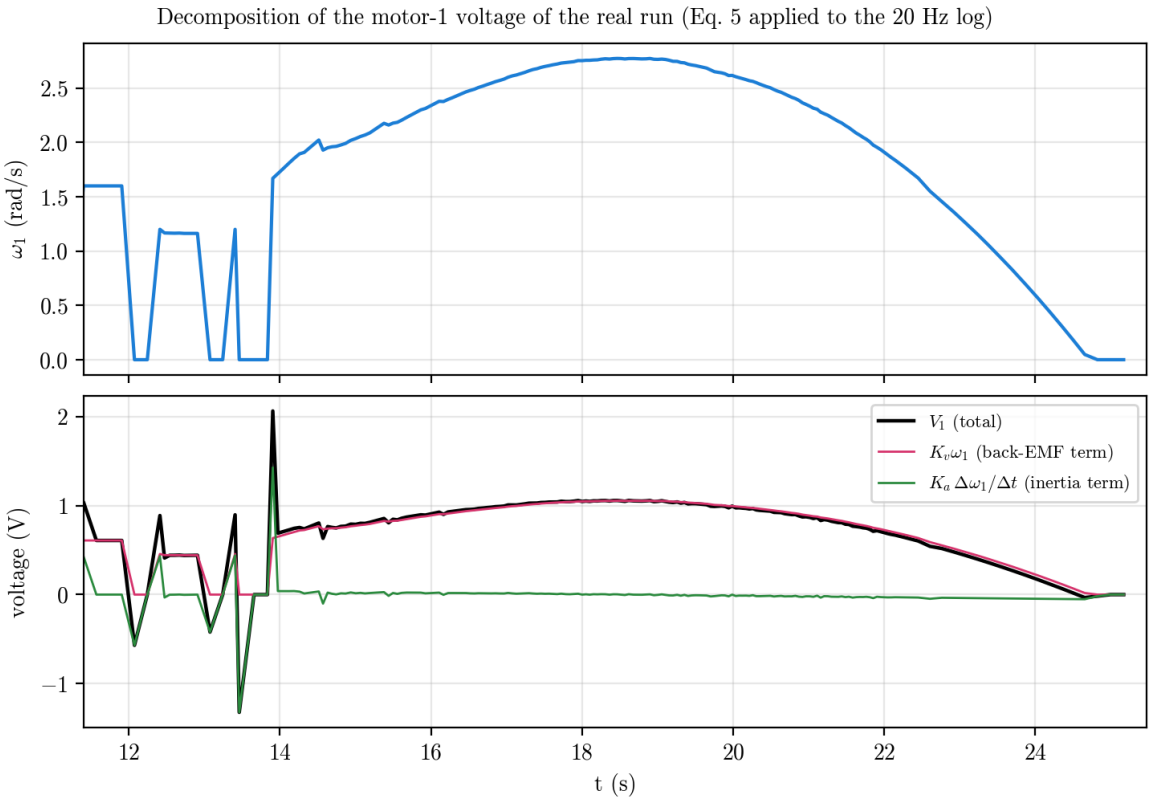



**Figure 9:** Back-EMF and inertia terms of $V_1$ in run 4 (20 Hz log).

### (iii) Live table and CSV logging

The node prints one table line per second and writes the same values to `motor_voltage_log.csv` (columns: `t_s`, `V1_V..V4_V`, `state`, `v_mps`, `w_radps`, `distance_m`, `bearing_rad`, `tof_m`, `w1..w4_radps`); a 20 Hz copy (`motor_voltage_fast.csv`) is used for the plots. When the state becomes ARRIVED the node waits three control ticks and then writes the last row at the true stop time, so that the voltages in that row are exactly zero.

flatness_nav console output (rendered from motor_voltage_log.csv)

```
ros2 launch rmf_flatness_nav flatness_nav.launch.py tag_id:=3 tag_size:=0.15 csv_path:=PWD/motor_voltage_log.csv
[INFO] [flatness_nav]: Tag tag36h11 id=3 size=0.15 m. Table header: t(s) V1 V2 V3 V4 state. CSV -> .../motor_voltage_log.…
[INFO] [flatness_nav]:     0 s | V1..V4 =   0.456  -0.456  -0.456   0.456 V | SEARCHING
[INFO] [flatness_nav]:     1 s | V1..V4 =   0.456  -0.456  -0.456   0.456 V | SEARCHING
[INFO] [flatness_nav]:     2 s | V1..V4 =   0.456  -0.456  -0.456   0.456 V | SEARCHING
[INFO] [flatness_nav]:     3 s | V1..V4 =   0.456  -0.456  -0.456   0.456 V | SEARCHING
[INFO] [flatness_nav]:     4 s | V1..V4 =   0.456  -0.456  -0.456   0.456 V | SEARCHING
[INFO] [flatness_nav]:     5 s | V1..V4 =   0.456  -0.456  -0.456   0.456 V | SEARCHING
[INFO] [flatness_nav]:     6 s | V1..V4 =   0.456  -0.456  -0.456   0.456 V | SEARCHING
[INFO] [flatness_nav]:     7 s | V1..V4 =   0.456  -0.456  -0.456   0.456 V | SEARCHING
[INFO] [flatness_nav]:     8 s | V1..V4 =   0.608  -0.608  -0.608   0.608 V | ALIGNING
[INFO] [flatness_nav]:     9 s | V1..V4 =   0.508  -0.508  -0.508   0.508 V | ALIGNING
[INFO] [flatness_nav]:    10 s | V1..V4 =   0.645   0.480   0.480   0.645 V | CHASING
[INFO] [flatness_nav]:    11 s | V1..V4 =   0.835   0.795   0.795   0.835 V | CHASING
[INFO] [flatness_nav]:    12 s | V1..V4 =   0.969   0.960   0.960   0.969 V | CHASING
[INFO] [flatness_nav]:    13 s | V1..V4 =   1.041   1.036   1.036   1.041 V | CHASING
[INFO] [flatness_nav]:    14 s | V1..V4 =   1.043   1.009   1.009   1.043 V | CHASING
[INFO] [flatness_nav]:    15 s | V1..V4 =   0.927   0.922   0.922   0.927 V | CHASING
[INFO] [flatness_nav]:    16 s | V1..V4 =   0.737   0.737   0.737   0.737 V | CHASING
[INFO] [flatness_nav]:    17 s | V1..V4 =   0.460   0.460   0.460   0.460 V | CHASING
[INFO] [flatness_nav]:    18 s | V1..V4 =   0.087   0.087   0.087   0.087 V | CHASING
[INFO] [flatness_nav]:  18.9 s | V1..V4 =   0.000   0.000   0.000   0.000 V | ARRIVED (trajectory end, tag out of view)
[INFO] [flatness_nav]: Halted at 18.9 s (trajectory end, tag out of view); log saved
```

**Figure 10:** Console output of the node for run 5, rendered from `motor_voltage_log.csv` in the node's log format (the console text itself was not saved during the run; the CSV is the real data).

**Voltage log of run 5 (flat tags).** The table below is Table 1 of the assignment, filled with the real values of run 5 (one row per second).

**Table 1.** Chassis motor armature voltage log (target: Tag 3), from `motor_voltage_log.csv`, one row per second.

| $t$ (s) | $V_1$ (V) | $V_2$ (V) | $V_3$ (V) | $V_4$ (V) | **State** |
|---|---|---|---|---|---|
| 0 | 0.456 | -0.456 | -0.456 | 0.456 | `SEARCHING` |
| 1 | 0.456 | -0.456 | -0.456 | 0.456 | `SEARCHING` |
| 2 | 0.456 | -0.456 | -0.456 | 0.456 | `SEARCHING` |
| 3 | 0.456 | -0.456 | -0.456 | 0.456 | `SEARCHING` |
| 4 | 0.456 | -0.456 | -0.456 | 0.456 | `SEARCHING` |
| 5 | 0.456 | -0.456 | -0.456 | 0.456 | `SEARCHING` |
| 6 | 0.456 | -0.456 | -0.456 | 0.456 | `SEARCHING` |
| 7 | 0.456 | -0.456 | -0.456 | 0.456 | `SEARCHING` |
| 8 | 0.608 | -0.608 | -0.608 | 0.608 | `ALIGNING` |
| 9 | 0.508 | -0.508 | -0.508 | 0.508 | `ALIGNING` |
| 10 | 0.645 | 0.480 | 0.480 | 0.645 | `CHASING` |
| 11 | 0.835 | 0.795 | 0.795 | 0.835 | `CHASING` |
| 12 | 0.969 | 0.960 | 0.960 | 0.969 | `CHASING` |
| 13 | 1.041 | 1.036 | 1.036 | 1.041 | `CHASING` |
| 14 | 1.043 | 1.009 | 1.009 | 1.043 | `CHASING` |
| 15 | 0.927 | 0.922 | 0.922 | 0.927 | `CHASING` |
| 16 | 0.737 | 0.737 | 0.737 | 0.737 | `CHASING` |
| 17 | 0.460 | 0.460 | 0.460 | 0.460 | `CHASING` |
| 18 | 0.087 | 0.087 | 0.087 | 0.087 | `CHASING` |
| $T_{\text{stop}}$=18.9 | 0.000 | 0.000 | 0.000 | 0.000 | `ARRIVED (trajectory end, tag out of view)` |

**Run 5 description** (tag `tag36h11`, ID 3, black square 0.15 m, printed on A4 and lying flat on the carpet with Tags 1 and 2 beside it as distractors; 360 p video; wired USB link; `logs/flat_run5/motor_voltage_log.csv`).

- **SEARCHING** from 0 to 8 s: the robot swept in place at 0.30 rad/s, so $V_1 = V_4 = +0.456$ V and $V_2 = V_3 = -0.456$ V.
- **ALIGNING** from 8 to 10 s: Tag 3 was first seen at a camera distance of 1.067 m and a bearing of 0.71 rad; the robot rotated in place until the bearing was below 0.18 rad (about 0.02 to 0.07 rad at the 10 s row).
- **CHASING** from 10 s: the cubic was planned from $d_0 = 1.053$ m, so $T = 1.6(1.053-0.20)/0.16 = 8.5$ s. The log reaches `ARRIVED` at 18.9 s, which is about 8.9 s after the first chasing row (1 s resolution). The peak speed is 0.136 m/s in the 1 Hz samples (the planner's own peak is 0.137 m/s). The camera distance fell from 1.053 to 0.465 m (at $t = 15$ s); after that the flat tag was no longer detected.
- **ARRIVED** at 18.9 s: the trajectory ended with the tag out of view, logged as `ARRIVED (trajectory end, tag out of view)`, with $V_1 = V_2 = V_3 = V_4 = 0.000$ V.

**Measured distances (tape measure).** The tags started about 1.2 to 1.5 m in front of the robot (a hand-measured range, not a single number) and the final gap between the front of the robot and the *edge of the A4 sheet* was **about 20 cm**; the sheet was not touched. On the printed sheet the black square starts 105 mm above the sheet's lower edge and is 150 mm wide, so the same stop is about $20 + 10.5 = 30.5$ cm from the near edge of the black square and about $30.5 + 7.5 = 38$ cm from the tag centre (assuming the sheet lay square to the robot and was printed at 100 %). The planner stops when the camera-to-centre distance is 0.20 m, so in run 5 the robot stopped about 18 cm *short* of the planned stop, never closer than planned. For comparison, the camera reported

1.067 m (to the tag centre) at the first detection. A planned travel of $1.053 - 0.20 = 0.85$ m from a true centre distance of about 1.23 m would leave about 0.38 m, which is what was measured; this suggests that the camera range of a flat tag read low by roughly 0.15 to 0.2 m (the camera tilt of $20° \pm 3°$ and its height are estimates), but with a hand-measured start range this explanation is not verified. The stop itself is dead-reckoned from the planned trajectory (the flat tag was no longer visible during the last part of the approach).

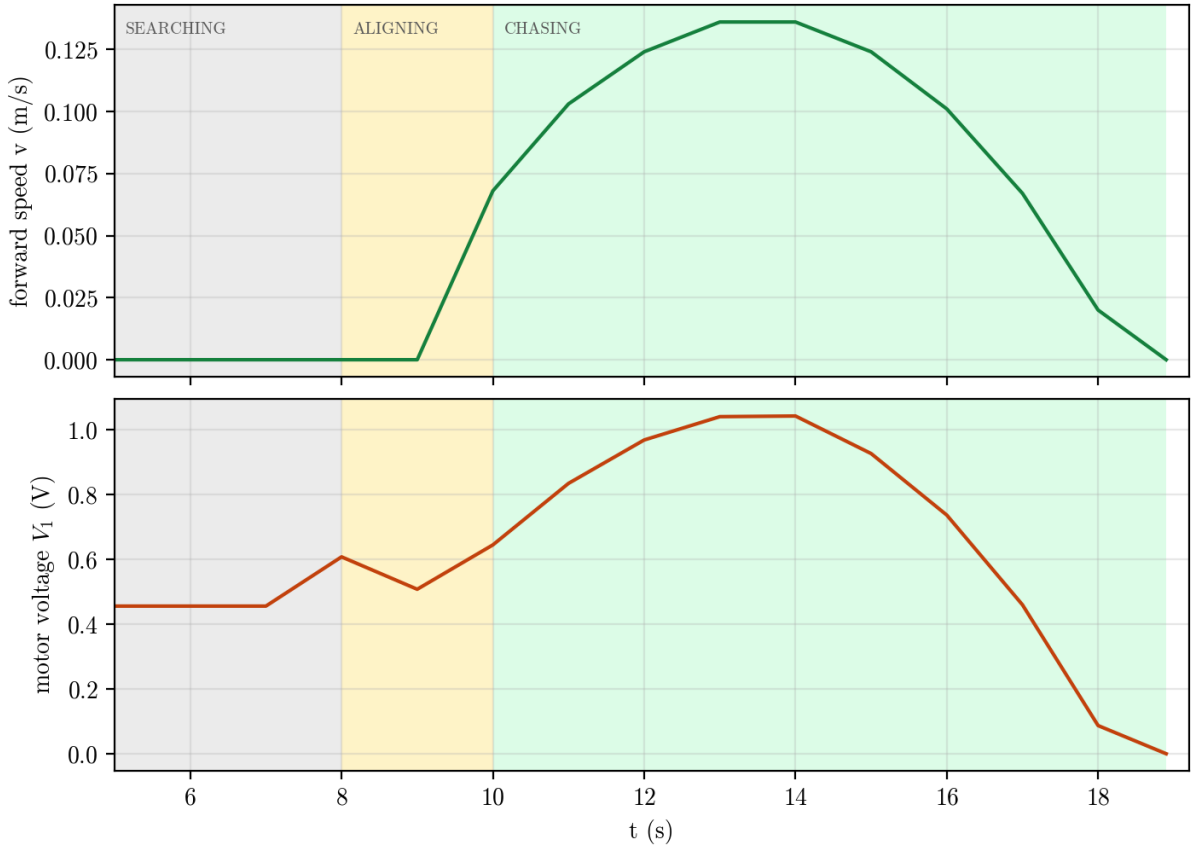



**Figure 11:** Forward speed $v(t)$ and motor voltage $V_1(t)$ during run 5 (flat tags, 1 Hz log; the time axis starts at 5 s, the search before it has $v = 0$ and $V_1 = 0.456$ V).

**Run 4 for comparison (upright tag, 20 Hz log).** (tag `tag36h11`, ID 3, black square 0.15 m, standing upright facing the robot; 360 p video; wired USB link; `motor_voltage_log.csv`).

- **SEARCHING** from 0 to 11.4 s: the robot started facing away from the tag and swept in place at 0.30 rad/s, so $V_1 = V_4 = +0.456$ V and $V_2 = V_3 = -0.456$ V.
- **ALIGNING** from 11.4 to 13.9 s: the tag came into view and the robot rotated in place (alternating voltage signs while it corrected) until $|\theta| \leq 0.18$ rad, then settled for 0.3 s.
- **CHASING** from 13.9 to 24.8 s: the cubic trajectory was planned from $d_0 \approx 1.28$ m (bearing 0.15 rad), giving $T = 1.6(1.28 - 0.20)/0.16 = 10.8$ s, and the chase took 10.9 s. The peak speed was 0.137 m/s (limit 0.16). The measured camera distance fell from 1.280 m to 0.351 m (at $t = 22$ s) with a bearing error of about 0.01 rad.
- **ARRIVED** at 24.8 s: below about 0.35 m the upright tag left the top of the image (its centre is above the optical axis), so the last part of the approach ran from the planned trajectory alone, which ends at the 20 cm standoff. The state was logged as `ARRIVED (trajectory end, tag out of view)`. The row at $T_{\text{stop}} = 25.2$ s shows $V_1 = V_2 = V_3 = V_4 = 0.000$ V, as required.

**Measured stop distance:** for run 4, a tape measure gave a gap of **20 cm** between the front of the robot and the tag after it halted, equal to the target. An earlier trial with the same tag and

code (a longer 36 s search) also gave 20 cm. Measurements are by hand (resolution about 1 cm) and the stop is reached by the dead-reckoned trajectory, so it depends on the distance estimate taken at the start of the chase.

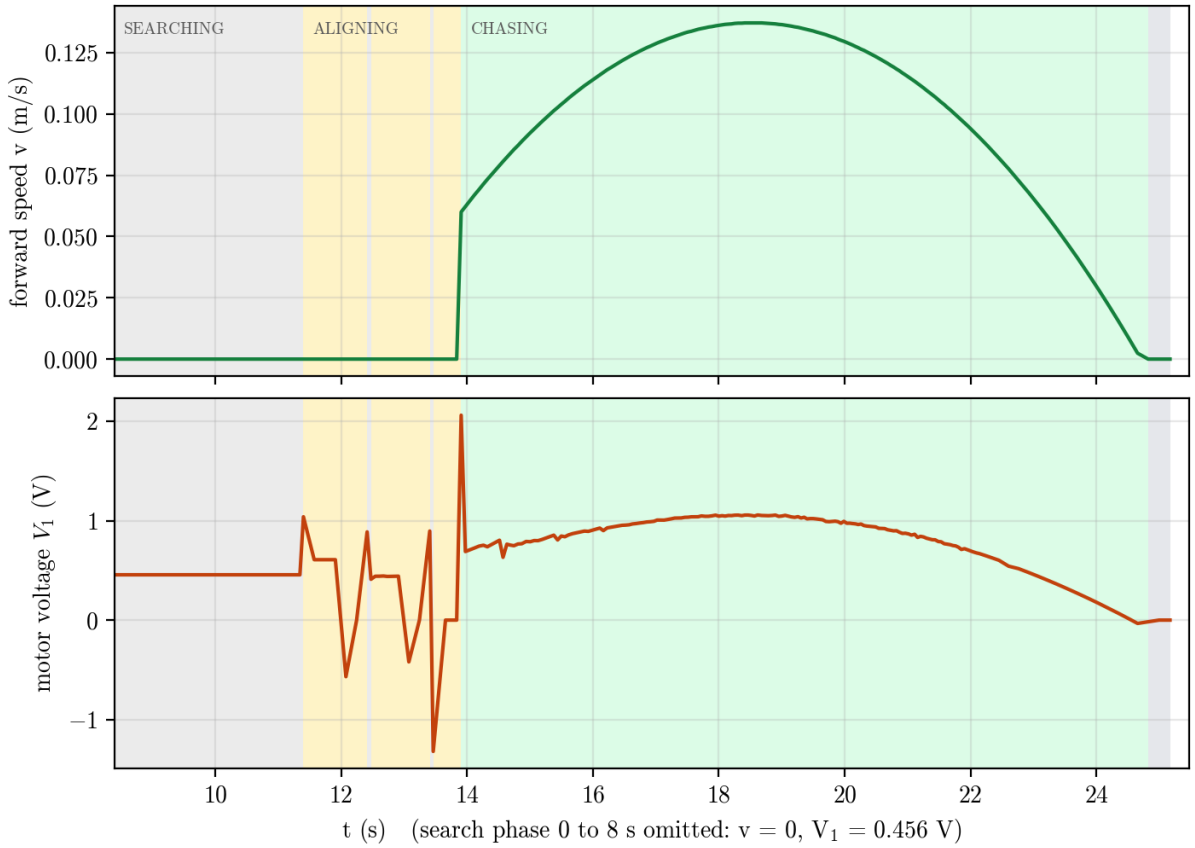



**Figure 12:** Run 4 (upright tag): $v(t)$ and $V_1(t)$ from the 20 Hz log, for comparison with the 1 Hz log of run 5.

**Repeatability.** Aligned at the start of the chase, the two upright-tag runs (3 and 4) are almost the same (Figure 13): the peak speed is 0.137 m/s in both, and the chase lasted 11.8 s and 10.9 s for start distances of 1.37 m and 1.28 m. The run-history table in Section 1.4 lists all four runs.

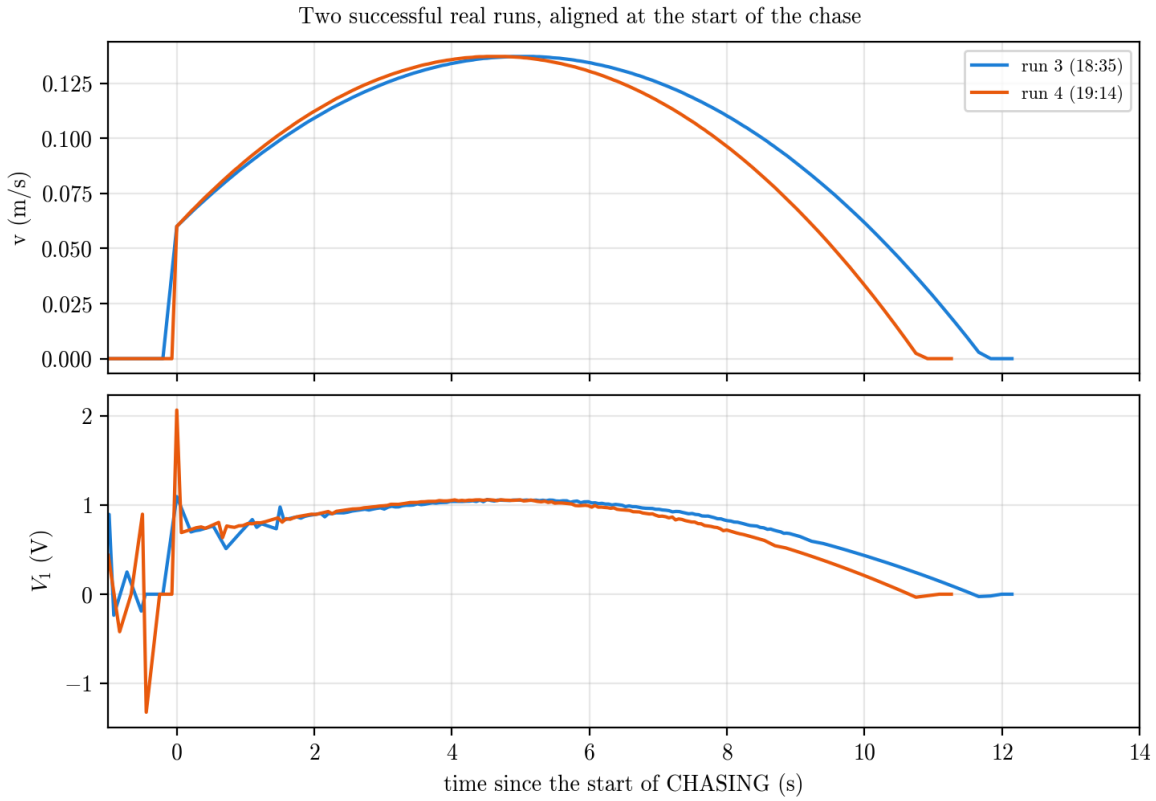



**Figure 13:** Runs 3 and 4 aligned at the start of CHASING (real data, 20 Hz log).

## Verification

**Unit tests.** 18 tests cover the boundary conditions of Eq. (2), the horizon formula and its 4 s floor, zero turn rate and the speed limit of the straight trajectory, Eq. (1) on a circle and at standstill, Eq. (4) and Eq. (5) values, every state transition (including the brake, loss before the chase, and continuing after loss during the chase), and the detector's recovery of range and bearing from rendered tags with the real camera intrinsics.

```
ameen@ameen-G6: ~/ws  (unit tests)

$ python -m pytest -v src/rm_flatness_nav/test
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-7.4.4, pluggy-1.4.0 -- /home/ameen/rm_venv/bin/python
rootdir: /home/ameen/robomasters/assignment2/ws/src/rm_flatness_nav
plugins: colcon-core-0.21.3, cov-4.1.0
collecting ... collected 18 items

test/test_core.py::test_polynomial_meets_boundary_conditions PASSED [  5%]
test/test_core.py::test_time_horizon_formula_and_floor PASSED [ 11%]
test/test_core.py::test_straight_trajectory_has_zero_turn_rate_and_limited_speed PASSED [ 16%]
test/test_core.py::test_flat_inversion_circle PASSED [ 22%]
test/test_core.py::test_flat_inversion_standstill_is_finite PASSED [ 27%]
test/test_core.py::test_wheel_rates_eq4 PASSED       [ 33%]
test/test_core.py::test_voltage_model_eq5 PASSED     [ 38%]
test/test_core.py::test_fsm_search_turns_toward_last_seen_side PASSED [ 44%]
test/test_core.py::test_fsm_aligns_in_place_before_chasing PASSED [ 50%]
test/test_core.py::test_fsm_brake_on_camera_and_tof PASSED [ 55%]
test/test_core.py::test_fsm_chase_speed_within_limit PASSED [ 61%]
test/test_core.py::test_fsm_loss_before_chase_returns_to_search PASSED [ 66%]
test/test_core.py::test_fsm_tag_leaving_image_during_chase_keeps_following_trajectory PASSED [ 72%]
test/test_core.py::test_fsm_blind_chase_can_be_disabled PASSED [ 77%]
test/test_detector.py::test_detects_and_recovers_ground_range_and_bearing[0.7-0.0] PASSED [ 83%]
test/test_detector.py::test_detects_and_recovers_ground_range_and_bearing[0.7-0.1] PASSED [ 88%]
test/test_detector.py::test_detects_and_recovers_ground_range_and_bearing[0.7--0.1] PASSED [ 94%]
test/test_detector.py::test_ignores_other_tag_ids PASSED [100%]

============================== 18 passed in 0.71s ==============================
```

**Figure 14:** Output of `pytest -v` (real, run on the host).

```
ameen@ameen-G6: ~/ws  (colcon build)

$ cd ws && colcon build --symlink-install --packages-select rm_flatness_nav
Starting >>> rm_flatness_nav
Finished <<< rm_flatness_nav [4.14s]

Summary: 1 package finished [4.48s]
```

**Figure 15:** The package builds with `colcon build -symlink-install`.

**Offline simulation (not hardware data).** The same state machine, trajectory and voltage model were simulated against a point-robot model with a tag that becomes invisible inside a minimum range, to exercise cases that were not run on the robot (different starting headings). All three scenarios halt at the standoff.

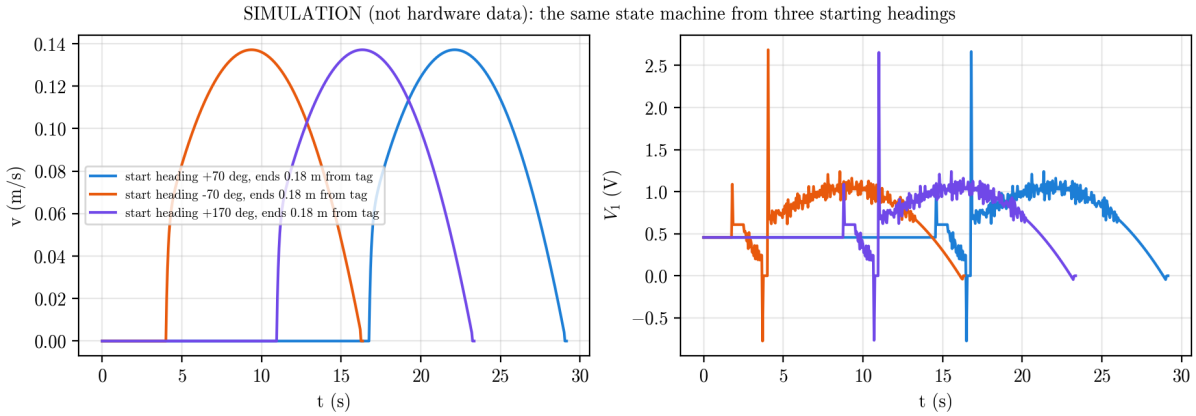



**Figure 16:** Simulation only: three starting headings.

```
ameen@ameen-G6: ~/ws  (offline simulation, not hardware data)

$ python -m rm_flatness_nav.sim --distance 1.0 --lateral 0.1 --heading-deg 60 --detect-range 1.2 --min-range 0.6   # …
SIMULATION: final state ARRIVED (trajectory end, tag out of view), distance to tag 0.183 m
         0       0.456      -0.456      -0.456       0.456   SEARCHING
         1       0.456      -0.456      -0.456       0.456   SEARCHING
         2       0.456      -0.456      -0.456       0.456   SEARCHING
         3       0.456      -0.456      -0.456       0.456   SEARCHING
         4       0.456      -0.456      -0.456       0.456   SEARCHING
         5       0.456      -0.456      -0.456       0.456   SEARCHING
       ...         ...         ...         ...         ...         ...
        25       0.236       0.236       0.236       0.236     CHASING
      25.9       0.000       0.000       0.000       0.000  ARRIVED (trajectory end, tag out of view)
```

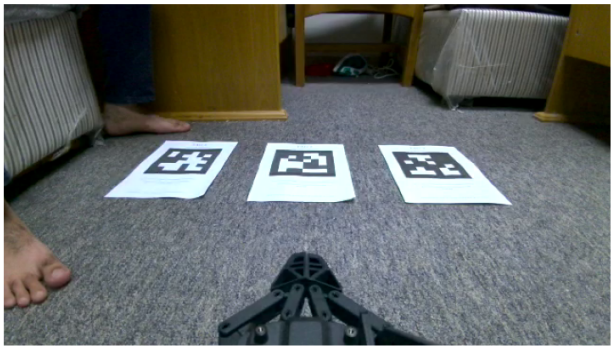
**Figure 17:** Output of `python -m rm_flatness_nav.sim` (simulation, not hardware data).

## Run 5 and the trials with the printed tags lying flat

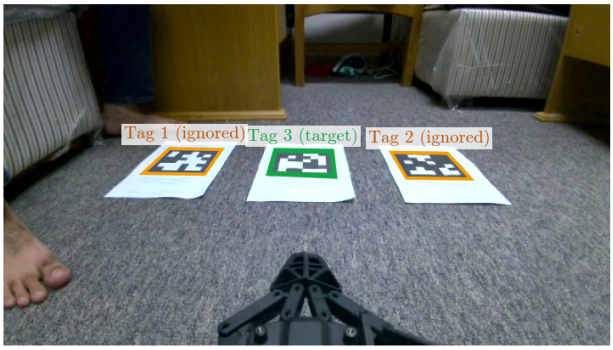
After the runs above, all three printed tags (`tag36h11`, IDs 1, 2, 3, black square 150 mm) were laid flat on the carpet, as the assignment describes, with Tag 3 in the middle and Tags 1 and 2 about 0.3 m to its left and right as distractors. The node follows only ID 3.

**Static detection test (real camera frames).** With the row 0.6 m in front of the robot, the project's own detector (native scale) found all three tags: Tag 1 83×27 px, Tag 3 69×27 px, Tag 2 84×28 px (Figure 18). With the row moved to about 0.8 m only Tag 2 was found, at 47×9 px, and Tag 3 was missed; at about 1 m nothing was found. So in a single frame a flat tag is readable out to roughly 0.6 to 0.7 m with this camera. The node's search is more tolerant (it also tries a 2× enlarged image and sees the tag over several frames).

(a) raw frame, robot camera



(b) detections of the project detector




(c) Tag 3 enlarged 6x: 69 x 27 px in the 640x360 frame; estimated ground range 0.71 m, bearing +0.8 deg (camera tilt 20 deg, tag 0.15 m)



**Figure 18:** What the robot's camera sees of Tag 3 (real frame, tags placed 0.6 m in front of the robot). (a) Raw frame; the claw of the gripper is visible at the bottom. (b) The three printed tags lie flat and all are found by the project's detector; only ID 3 (green) is the target, the others (orange) are ignored. (c) Tag 3 enlarged six times: the flat tag is only 27 px tall, yet its 6×6 data cells and black border are still readable. The ground range from the pose of this single frame is 0.71 m (the tags were placed at 0.6 m from the front of the robot, the range is measured from the camera to the tag centre).

**Mission attempts.** Four autonomous attempts were made with this layout. **One reached ARRIVED: run 5** (Table 1 and the main plot). In the first attempt the tag was not detected during a 33 s sweep. The other two (later) attempts detected the tag, aligned and chased for several seconds, then lost it and returned to SEARCHING without halting; in both the logged distances were about 100 times too large (for example 95.8 in the log, i.e. 0.96 m if the factor of 100 is removed), which is what a tag size typed as 15 instead of 0.15 would produce; because of this scale error the rule that continues the trajectory when the tag leaves the image (it only applies within 1.8 m) could not act. This cause is inferred from the numbers; the node's start-up line with the tag size was not kept. The tag-size prompt of the helper script now rejects implausible values. The logs of the unsuccessful attempts are in `logs/flat_failed_attempts/`. Run 5 has no 20 Hz log (only `motor_voltage_log.csv` at 1 Hz was written), so its plot has one point per second.

**Photographs of the trials.** Figure 19 shows phone photographs taken during the flat-tag trials. Panel (a) shows the starting layout with the phone's measure tool reading 1.5 m between the robot and the tags, which supports the upper end of the hand-measured start range of 1.2 to 1.5 m; panel (b) is a closer placement (69 cm) photographed during the trials, and is not used for any number in this report. Panels (c) and (d) show the robot after it stopped in front of Tags 1, 3 and 2 (left to right); the stop gap quoted in the text (about 20 cm to the sheet edge) is the hand measurement

and is not read off these perspective photos, but the photos agree that the front of the robot is a clear distance from the black square.

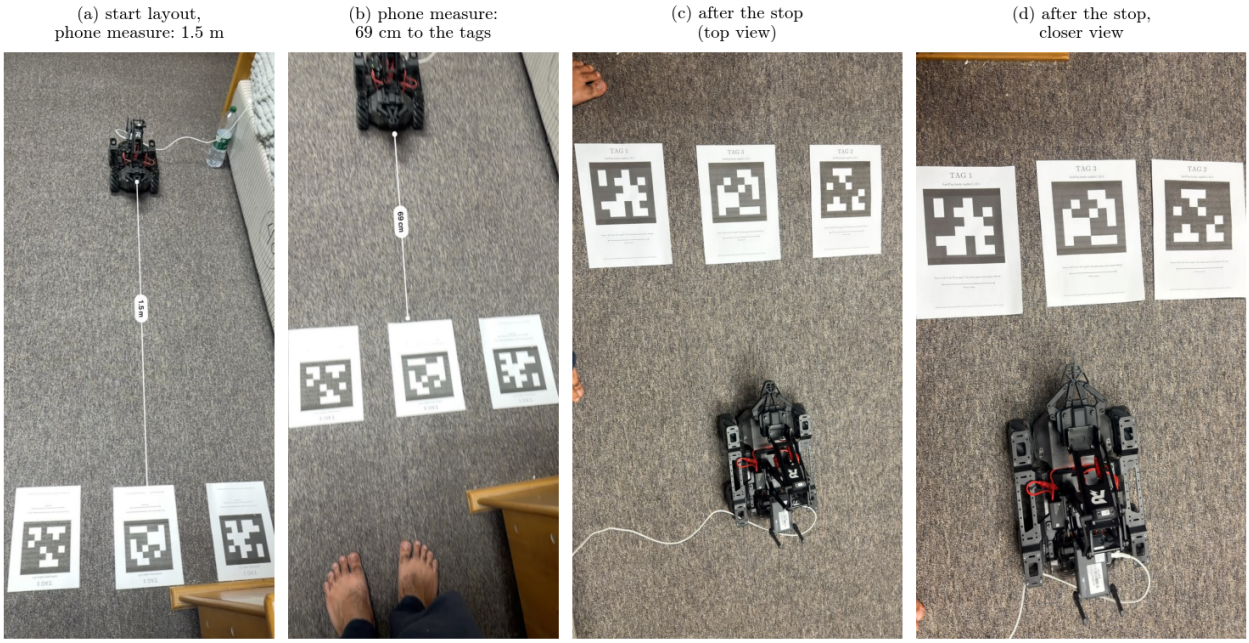



**Figure 19:** Phone photographs from above during the flat-tag trials: (a) start layout with the phone measure at 1.5 m, (b) a closer placement (69 cm), (c) and (d) the robot after it stopped.

## Deviations from the assignment text and limitations

- **Tag orientation.** The assignment places the tag *flat* on the floor. Run 5 does so. Runs 3 and 4 used the tag standing upright because a flat 15 cm tag is a thin sliver for this camera (Figure 6); a single frame reads a flat tag only out to about 0.6 to 0.7 m, although the node (with its upscaled search and several frames) found it at 1.07 m in run 5.
- **Distance.** The assignment gives 1.5 to 1.8 m. In run 5 the tags started about 1.2 to 1.5 m away by hand measurement (the camera reported 1.07 m to the tag centre at the first detection), at or slightly below that range, and the robot stopped about 20 cm from the sheet edge (about 38 cm from the tag centre) instead of the planned 20 cm. In runs 3 and 4: The tag was about 1.3 to 1.4 m away (the robot only rotates before the chase, so the camera range at the first detection is the starting distance: 1.375 m in run 3 and 1.28 m in run 4). The detector could see further with its upscaled search, but the tag was not placed there.
- **Camera and arm.** The arm pose did not change the camera view; the camera tilt of about 20° was read from images.
- **Stop source.** The stop came from the planned trajectory, not from a camera or ToF reading. The sensor brakes are implemented and unit-tested but were not triggered on the robot.
- **Voltages** are model values computed from commanded velocities (Eq. (5)); they are not measured at the motors.
- **Number of trials.** Three successful runs (3 and 4 upright, 5 flat); stop gaps were measured by hand to about 1 cm. Run 5 has only a 1 Hz log and a hand-measured start range.
- **Tags 1 and 2.** Runs 3 and 4 used Tag 3 alone. In run 5 and the other flat-tag trials, Tags 1 and 2 lay beside Tag 3 as distractors; the node followed only ID 3 and the log does not record which other tags were seen.

## Deliverables checklist

| Item | Status | Where |
|---|---|---|
| Equations for flatness inversion and the boundary matrix solution | done | Eq. (1), Eq. (2) |
| Completed Table 1 with experimental voltages | done | Section 4(iii) |
| Plot of $v(t)$ and $V_1(t)$ | done | Section 4(iii) |
| Code archive: package `rm_flatness_nav` and `motor_voltage_log.csv` | done | `assignment2_submission.zip`, repository |
| Demonstration video ($\leq$ 45 s): sweeping, centering, cruising and stopping at 20 cm | done | Video 1: `https://youtu.be/EaZKQfd3usE`; Video 2: `https://youtu.be/JlHnuVnLLrA` |

## Source listings

### `flatness.py`

```
"""Differential-flatness trajectory generation for a unicycle-type robot.

The flat output is the planar position z(t) = [x(t), y(t)]^T.  Every smooth curve
for z(t) gives the linear speed v and turn rate w in closed form (Eq. 1), so no
numerical optimiser is needed.  Each component is a cubic over horizon T (Eq. 2).
"""
import math

import numpy as np

EPS = 1e-6
V_MAX = 0.16        # m/s cruise limit (Eq. 3)
STANDOFF = 0.20     # m, final distance from the tag
T_MIN = 4.0         # s
T_GAIN = 1.6


def flat_inversion(xd, yd, xdd, ydd):
    """Eq. (1): recover (v, w) from the first and second derivatives of z(t)."""
    speed_sq = xd * xd + yd * yd
    v = math.sqrt(speed_sq)
    if speed_sq < EPS:
        # heading is undefined at standstill (0/0): command no turning
        return v, 0.0
    w = (xd * ydd - yd * xdd) / speed_sq
    return v, w


def solve_flat_polynomial(z0, zdot0, zf, zdotf, T):
    """Eq. (2): cubic coefficients [a0, a1, a2, a3] of z(t) matching the four
    boundary values z(0), z'(0), z(T), z'(T).  Solved with numpy.linalg.solve."""
    A = np.array([[1.0, 0.0, 0.0, 0.0],
                  [0.0, 1.0, 0.0, 0.0],
                  [1.0, T, T ** 2, T ** 3],
                  [0.0, 1.0, 2.0 * T, 3.0 * T ** 2]])
    b = np.array([z0, zdot0, zf, zdotf], dtype=float)
    return np.linalg.solve(A, b)


def eval_cubic(alpha, t):
    """Position, velocity and acceleration of the cubic at time t."""
    a0, a1, a2, a3 = alpha
    return (a0 + a1 * t + a2 * t ** 2 + a3 * t ** 3,
            a1 + 2.0 * a2 * t + 3.0 * a3 * t ** 2,
            2.0 * a2 + 6.0 * a3 * t)


def time_horizon(d0, standoff=STANDOFF, v_max=V_MAX, t_min=T_MIN, gain=T_GAIN):
    """Eq. (3): T = max(4.0, 1.6 (d0 - 0.20) / v_max)."""
    return max(t_min, gain * (d0 - standoff) / v_max)


class FlatTrajectory:
    """Straight-ahead cubic pursuit in the robot frame at the moment of centering.

    Boundary conditions (assignment, Problem 2 ii):
        x(0)=0, x'(0)=0.06 m/s, x(T)=d0-0.20, x'(T)=0;  y(0)=0, y'(0)=0, y(T)=0, y'(T)=0.
    """

    def __init__(self, d0, v0=0.06, standoff=STANDOFF, v_max=V_MAX):
        self.d0 = d0
        self.T = time_horizon(d0, standoff, v_max)
        goal = max(d0 - standoff, 0.0)
        self.alpha_x = solve_flat_polynomial(0.0, v0, goal, 0.0, self.T)
        self.alpha_y = solve_flat_polynomial(0.0, 0.0, 0.0, 0.0, self.T)

    def sample(self, t):
        """(v, w, x, y) at time t since the trajectory started; clamps to [0, T]."""
        t = min(max(t, 0.0), self.T)
        x, xd, xdd = eval_cubic(self.alpha_x, t)
```

```
        y, yd, ydd = eval_cubic(self.alpha_y, t)
        v, w = flat_inversion(xd, yd, xdd, ydd)
        # the cubic may undershoot slightly below zero speed near T; never reverse
        if xd < 0.0:
            v = 0.0
        return v, w, x, y

    def finished(self, t):
        return t >= self.T
```

## wheel_voltage.py

```
"""Wheel-rate kinematics (Eq. 4) and armature voltage model (Eq. 5) for the RoboMaster EP."""

WHEEL_RADIUS = 0.05   # r, m
CHASSIS_RADIUS = 0.20  # L_c, m
K_V = 0.38            # V.s/rad  (back-EMF)
K_A = 0.06            # V.s^2/rad (inertia)


def wheel_rates(vx, wz, r=WHEEL_RADIUS, lc=CHASSIS_RADIUS):
    """Eq. (4): returns (w1, w2, w3, w4) = (w_FR, w_FL, w_RL, w_RR) in rad/s."""
    w1 = (vx + lc * wz) / r   # front-right
    w2 = (vx - lc * wz) / r   # front-left
    return (w1, w2, w2, w1)   # w3 = w_RL = w2, w4 = w_RR = w1


class MotorVoltageModel:
    """Eq. (5): V_i = K_v w_i + K_a (w_i(t) - w_i(t-dt)) / dt, per motor."""

    def __init__(self, kv=K_V, ka=K_A):
        self.kv = kv
        self.ka = ka
        self._prev = None

    def reset(self):
        self._prev = None

    def step(self, omegas, dt):
        """Voltages for the current wheel rates; dt is the time since the last call."""
        if self._prev is None or dt <= 0.0:
            accel = (0.0,) * len(omegas)
        else:
            accel = tuple((w - p) / dt for w, p in zip(omegas, self._prev))
        self._prev = tuple(omegas)
        return tuple(self.kv * w + self.ka * a for w, a in zip(omegas, accel))
```

## state_machine.py

```
"""Supervisory state machine: SEARCHING -> ALIGNING -> CHASING -> ARRIVED.

Pure logic (no ROS), driven by `update()` once per control tick, so it can be unit
tested and simulated.  Sign convention: bearing > 0 means the tag is to the LEFT of the
camera axis (ROS REP-103, +w = counter-clockwise), so aligning is w = +K * bearing.
"""
from dataclasses import dataclass
from typing import Optional

from .flatness import FlatTrajectory

SEARCHING, ALIGNING, CHASING, ARRIVED = 'SEARCHING', 'ALIGNING', 'CHASING', 'ARRIVED'


@dataclass
class Params:
    lost_timeout: float = 0.45       # s without a tag -> SEARCHING
    search_rate: float = 0.30        # rad/s, in-place sweep
    align_tol: float = 0.18          # rad (about 10 deg) to start chasing
    realign_tol: float = 0.30        # rad, hysteresis: leave CHASING only beyond this
    settle_time: float = 0.30        # s the bearing must stay inside align_tol
    align_gain: float = 0.8          # 1/s
    align_min_rate: float = 0.12     # rad/s, overcome wheel dead-band
    align_max_rate: float = 0.40     # rad/s
    chase_bearing_gain: float = 0.8  # 1/s, small correction added to the flat-output w
    chase_max_w: float = 0.40        # rad/s
    brake_distance: float = 0.20     # m, camera distance or ToF threshold
    v_max: float = 0.16              # m/s
    blind_chase: bool = True         # keep following the trajectory if the tag leaves the image
    blind_max_range: float = 1.8     # m, only if the tag was last seen at most this far away


@dataclass
class Observation:
    now: float
    visible: bool = False                # tag detected in the latest frame
    bearing: float = 0.0                 # rad, valid when visible
    distance: float = float('inf')       # m, ground-plane camera distance, valid when visible
    tof: Optional[float] = None          # m, front ToF reading (None if unavailable)


def _clip(x, lo, hi):
    return max(lo, min(hi, x))


class NavStateMachine:
    def __init__(self, params: Params = None):
```

```
        self.p = params or Params()
        self.state = SEARCHING
        self.last_seen = None            # time of the last sighting
        self.last_dir = 1.0              # +1 CCW (tag was left), -1 CW
        self.centered_since = None
        self.traj = None
        self.traj_t0 = None
        self.arrival_reason = ''
        self.last_distance = float('inf')

    # -- helpers ---------------------------------------------------------------
    def _brake_hit(self, obs):
        if obs.visible and obs.distance <= self.p.brake_distance:
            return 'camera'
        if obs.tof is not None and obs.tof <= self.p.brake_distance:
            return 'tof'
        return ''

    def _enter_search(self):
        self.state = SEARCHING
        self.traj = None
        self.centered_since = None

    # -- main step -------------------------------------------------------------
    def update(self, obs: Observation):
        """Returns (v, w) to command this tick; self.state holds the current state."""
        p = self.p
        if self.state == ARRIVED:
            return 0.0, 0.0

        if obs.visible:
            self.last_seen = obs.now
            self.last_distance = obs.distance
            if abs(obs.bearing) > 1e-3:
                self.last_dir = 1.0 if obs.bearing > 0 else -1.0

        # active brake: engages immediately from any state
        reason = self._brake_hit(obs)
        if reason:
            self.state, self.arrival_reason, self.traj = ARRIVED, reason, None
            return 0.0, 0.0

        lost = self.last_seen is None or (obs.now - self.last_seen) > p.lost_timeout
        # A floor tag leaves the image when the robot is close (the camera cannot look straight down
        # at its own feet).  While chasing, keep following the planned trajectory without the tag:
        # it already ends at the 20 cm standoff.
        if (lost and p.blind_chase and self.state == CHASING and self.traj is not None
                and self.last_distance <= p.blind_max_range):
            return self._chase(obs)
        if lost:
            if self.state != SEARCHING:
                self._enter_search()
            return 0.0, p.search_rate * self.last_dir

        # the tag is (recently) visible
        if self.state == SEARCHING:
            self.state = ALIGNING
            self.centered_since = None

        if self.state == ALIGNING:
            if not obs.visible:
                return 0.0, 0.0          # between frames: hold still, do not guess
            if abs(obs.bearing) <= p.align_tol:
                if self.centered_since is None:
                    self.centered_since = obs.now
                if obs.now - self.centered_since >= p.settle_time:
                    self.traj = FlatTrajectory(obs.distance, v_max=p.v_max)
                    self.traj_t0 = obs.now
                    self.state = CHASING
                    return self._chase(obs)
                return 0.0, 0.0          # settling, in place
            self.centered_since = None
            w = _clip(p.align_gain * obs.bearing, -p.align_max_rate, p.align_max_rate)
            if abs(w) < p.align_min_rate:
                w = p.align_min_rate * (1.0 if w >= 0 else -1.0)
            return 0.0, w

        # CHASING
        if obs.visible and abs(obs.bearing) > p.realign_tol:
            self.state, self.traj, self.centered_since = ALIGNING, None, None
            return 0.0, 0.0
        return self._chase(obs)

    def _chase(self, obs):
        t = obs.now - self.traj_t0
        if self.traj.finished(t):
            self.state, self.traj = ARRIVED, None
            self.arrival_reason = 'trajectory end' if obs.visible else 'trajectory end, tag out of view'
            return 0.0, 0.0
        v, w_ff, _, _ = self.traj.sample(t)
        w = w_ff
        if obs.visible:
            w += self.p.chase_bearing_gain * obs.bearing
        return min(v, self.p.v_max), _clip(w, -self.p.chase_max_w, self.p.chase_max_w)
```

## tag_detector.py

```
"""AprilTag detection (OpenCV aruco AprilTag dictionaries) and floor-plane geometry."""
import math
```

```
from typing import Optional

import cv2
import numpy as np

FAMILIES = {
    'tag16h5': 'DICT_APRILTAG_16h5', 'tag25h9': 'DICT_APRILTAG_25h9',
    'tag36h10': 'DICT_APRILTAG_36h10', 'tag36h11': 'DICT_APRILTAG_36h11',
}

# optical frame (x right, y down, z forward) -> base frame (x forward, y left, z up)
R_OPTICAL_TO_BASE = np.array([[0.0, 0.0, 1.0],
                              [-1.0, 0.0, 0.0],
                              [0.0, -1.0, 0.0]])


def camera_rotation(pitch_down):
    """Rotation optical -> base for a camera tilted down by `pitch_down` rad."""
    c, s = math.cos(pitch_down), math.sin(pitch_down)
    ry = np.array([[c, 0.0, s], [0.0, 1.0, 0.0], [-s, 0.0, c]])  # +y rotation tilts +x down
    return ry @ R_OPTICAL_TO_BASE


def ground_range_bearing(p_cam, r_base_cam):
    """Horizontal distance and bearing of a point given in the camera optical frame.

    r_base_cam rotates optical vectors into the base frame.  Bearing > 0 = target on the LEFT.
    Range is measured from the camera, projected onto the floor plane.
    """
    p = r_base_cam @ np.asarray(p_cam, dtype=float)
    return math.hypot(p[0], p[1]), math.atan2(p[1], p[0])


class TagDetection:
    def __init__(self, tag_id, corners, tvec, rvec):
        self.tag_id, self.corners, self.tvec, self.rvec = tag_id, corners, tvec, rvec


class TagDetector:
    def __init__(self, family='tag36h11', tag_id=3, tag_size=0.10):
        name = FAMILIES[family]
        self.tag_id, self.size = tag_id, tag_size
        self._dict = cv2.aruco.getPredefinedDictionary(getattr(cv2.aruco, name))
        # OpenCV 4.6 (Ubuntu 24.04 / ROS Jazzy) segfaults when a DetectorParameters() built with
        # the constructor is modified; its _create() factory is safe.  4.7+ only has the constructor.
        self._params = (cv2.aruco.DetectorParameters_create()
                        if hasattr(cv2.aruco, 'DetectorParameters_create')
                        else cv2.aruco.DetectorParameters())
        self._params.cornerRefinementMethod = cv2.aruco.CORNER_REFINE_APRILTAG
        self._new_api = hasattr(cv2.aruco, 'ArucoDetector')
        if self._new_api:
            self._detector = cv2.aruco.ArucoDetector(self._dict, self._params)
        s = tag_size / 2.0
        self._obj = np.array([[-s, s, 0], [s, s, 0], [s, -s, 0], [-s, -s, 0]], dtype=np.float32)

    def detect(self, gray, camera_matrix, dist_coeffs=None, scales=(1.0, 2.0)) -> Optional[TagDetection]:
        """Pose of the configured tag id, or None if it is not in the image.

        A small tag (about 30 px per side or less, i.e. a 15 cm tag beyond ~1.3 m at 360p) is often missed
        at native size but found after enlarging the image, so each scale is tried in order and only the
        first success is used.  Corners are mapped back to native pixels before the pose is solved."""
        for scale in scales:
            img = gray if scale == 1.0 else cv2.resize(gray, None, fx=scale, fy=scale, interpolation=cv2.INTER_CUBIC)
            det = self._detect_once(img, camera_matrix, dist_coeffs, scale)
            if det is not None:
                return det
        return None

    def _detect_once(self, gray, camera_matrix, dist_coeffs, scale):
        if self._new_api:
            corners, ids, _ = self._detector.detectMarkers(gray)
        else:
            corners, ids, _ = cv2.aruco.detectMarkers(gray, self._dict, parameters=self._params)
        if ids is None:
            return None
        for c, i in zip(corners, ids.flatten()):
            if self.tag_id is not None and int(i) != self.tag_id:
                continue
            img = (c.reshape(4, 2) / scale).astype(np.float32)
            dist = np.zeros(5) if dist_coeffs is None else dist_coeffs
            ok, rvec, tvec = cv2.solvePnP(self._obj, img, camera_matrix, dist, flags=cv2.SOLVEPNP_IPPE_SQUARE)
            if ok:
                return TagDetection(int(i), img, tvec.reshape(3), rvec.reshape(3))
        return None
```

## nav_node.py

```
"""ROS 2 node: AprilTag search, in-place alignment, flat-output pursuit, standoff brake,
and 1 Hz motor-voltage logging for the RoboMaster EP.

Subscribes : camera/image_color, camera/camera_info (sensor_msgs), range_0 (ToF, optional), TF
Publishes  : cmd_vel (geometry_msgs/Twist), target_arm_position (geometry_msgs/Point, at start)
"""
import csv
import math
import os
import threading
import time

```

```
import cv2
import numpy as np
import rclpy
from cv_bridge import CvBridge
from geometry_msgs.msg import Point, Twist
from rclpy.node import Node
from rclpy.qos import qos_profile_sensor_data
from sensor_msgs.msg import CameraInfo, Image, Range
import tf2_ros

from .state_machine import ARRIVED, NavStateMachine, Observation, Params
from .tag_detector import TagDetector, camera_rotation, ground_range_bearing
from .wheel_voltage import MotorVoltageModel, wheel_rates

CSV_HEADER = ['t_s', 'V1_V', 'V2_V', 'V3_V', 'V4_V', 'state',
              'v_mps', 'w_radps', 'distance_m', 'bearing_rad', 'tof_m',
              'w1_radps', 'w2_radps', 'w3_radps', 'w4_radps']


def quat_to_matrix(q):
    x, y, z, w = q.x, q.y, q.z, q.w
    return np.array([[1 - 2 * (y * y + z * z), 2 * (x * y - z * w), 2 * (x * z + y * w)],
                     [2 * (x * y + z * w), 1 - 2 * (x * x + z * z), 2 * (y * z - x * w)],
                     [2 * (x * z - y * w), 2 * (y * z + x * w), 1 - 2 * (x * x + y * y)]])


class GrabThread(threading.Thread):
    """Optional direct OpenCV capture (camera_source:=cv2).  Implements the assignment hint:
    a short sleep between cap.grab() calls lets the decoder recycle its buffers."""

    def __init__(self, url):
        super().__init__(daemon=True)
        self.cap, self.frame, self.lock, self.running = cv2.VideoCapture(url), None, threading.Lock(), True

    def run(self):
        while self.running:
            if self.cap.grab():
                ok, frame = self.cap.retrieve()
                if ok:
                    with self.lock:
                        self.frame = frame
            time.sleep(0.010)

    def latest(self):
        with self.lock:
            return self.frame


class FlatnessNav(Node):
    def __init__(self):
        super().__init__('flatness_nav')
        d = self.declare_parameter
        self.tag_family = d('tag_family', 'tag36h11').value
        self.tag_id = d('tag_id', 3).value
        self.tag_size = d('tag_size', 0.10).value          # metres, printed black-square side: MEASURE IT
        self.image_topic = d('image_topic', 'camera/image_color').value
        self.info_topic = d('info_topic', 'camera/camera_info').value
        self.tof_topic = d('tof_topic', 'range_0').value
        self.use_tof = d('use_tof', True).value
        self.base_frame = d('base_frame', 'base_link').value
        self.camera_frame = d('camera_frame', 'camera_optical_link').value
        # The camera is fixed to the chassis (verified from images), so the arm/URDF TF is NOT used by default.
        self.fallback_pitch_deg = d('camera_pitch_deg', 20.0).value   # measured: about 17-23 deg below horizontal
        self.use_tf = d('use_tf_camera', False).value
        self.control_rate = d('control_rate', 20.0).value
        self.csv_path = d('csv_path', 'motor_voltage_log.csv').value
        self.fast_path = d('fast_csv_path', 'motor_voltage_fast.csv').value   # 20 Hz log for the v(t), V1(t) plot
        self.set_arm = d('set_arm_pose', True).value
        self.arm_x, self.arm_z = d('arm_x_m', 0.105).value, d('arm_z_m', 0.030).value
        self.camera_source = d('camera_source', 'ros').value          # 'ros' or 'cv2'
        self.camera_url = d('camera_url', '').value
        self.detect_fresh = d('detection_fresh_s', 0.20).value
        fsm_params = Params()
        for name in Params.__dataclass_fields__:
            setattr(fsm_params, name, d(f'fsm.{name}', getattr(fsm_params, name)).value)
        self.fsm = NavStateMachine(fsm_params)
        self.voltage = MotorVoltageModel(d('kv', 0.38).value, d('ka', 0.06).value)
        self.r = d('wheel_radius', 0.05).value
        self.lc = d('chassis_radius', 0.20).value

        self.detector = TagDetector(self.tag_family, self.tag_id, self.tag_size)
        self.bridge = CvBridge()
        self.K, self.dist = None, None
        self.det_time, self.bearing, self.distance = -1e9, 0.0, float('inf')
        self.tof = None
        self.r_bc, self.tf_ok = camera_rotation(math.radians(self.fallback_pitch_deg)), False
        self.tf_buffer = tf2_ros.Buffer()
        self.tf_listener = tf2_ros.TransformListener(self.tf_buffer, self)
        self.grab = None

        self.cmd_pub = self.create_publisher(Twist, 'cmd_vel', 10)
        self.arm_pub = self.create_publisher(Point, 'target_arm_position', 1)
        self.create_subscription(CameraInfo, self.info_topic, self.on_info, 1)
        if self.camera_source == 'ros':
            self.create_subscription(Image, self.image_topic, self.on_image, qos_profile_sensor_data)
        else:
            self.grab = GrabThread(self.camera_url)
            self.grab.start()
        if self.use_tof:
            self.create_subscription(Range, self.tof_topic, self.on_range, qos_profile_sensor_data)

```

```
        self.t_start = self.now()
        self.prev_tick = self.t_start
        self.next_log = 0.0
        self.volts, self.omegas = (0.0,) * 4, (0.0,) * 4
        self.v_cmd = self.w_cmd = 0.0
        self.stop_ticks = 0
        self.finished = False
        self._csv_file = open(self.csv_path, 'w', newline='')
        self._csv = csv.writer(self._csv_file)
        self._csv.writerow(CSV_HEADER)
        self._fast_file = open(self.fast_path, 'w', newline='')
        self._fast = csv.writer(self._fast_file)
        self._fast.writerow(['t_s', 'v_mps', 'w_radps', 'V1_V', 'V2_V', 'V3_V', 'V4_V', 'state'])
        self.create_timer(1.0 / self.control_rate, self.tick)
        if self.set_arm:
            self._arm_sends = 0
            self.create_timer(1.0, self.send_arm)
        self.get_logger().info(
            f'Tag {self.tag_family} id={self.tag_id} size={self.tag_size} m. '
            f'Table header: t(s) V1 V2 V3 V4 state. CSV -> {os.path.abspath(self.csv_path)}')
        if self.tag_size == 0.10:
            self.get_logger().warn('tag_size is the 0.10 m default: set it to the MEASURED tag side length.')

    # ---- helpers ------------------------------------------------------------------
    def now(self):
        return self.get_clock().now().nanoseconds * 1e-9

    def send_arm(self):
        """Arm pose (X=105 mm, Y=30 mm): the driver's second coordinate is the vertical `z`."""
        self.arm_pub.publish(Point(x=self.arm_x, y=0.0, z=self.arm_z))
        self._arm_sends += 1

    def on_info(self, msg):
        self.K = np.array(msg.k, dtype=float).reshape(3, 3)
        self.dist = np.array(msg.d, dtype=float) if len(msg.d) else None

    def on_range(self, msg):
        self.tof = None if msg.range >= msg.max_range - 1e-3 else float(msg.range)

    def update_camera_rotation(self):
        try:
            tf = self.tf_buffer.lookup_transform(self.base_frame, self.camera_frame, rclpy.time.Time())
            self.r_bc, self.tf_ok = quat_to_matrix(tf.transform.rotation), True
        except Exception:
            pass   # keep the fixed-pitch fallback

    def process_frame(self, frame):
        if self.K is None:
            return
        gray = frame if frame.ndim == 2 else cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
        det = self.detector.detect(gray, self.K, self.dist)
        if det is None:
            return
        if self.use_tf:
            self.update_camera_rotation()
        self.distance, self.bearing = ground_range_bearing(det.tvec, self.r_bc)
        self.det_time = self.now()

    def on_image(self, msg):
        try:
            self.process_frame(self.bridge.imgmsg_to_cv2(msg, 'bgr8'))
        except Exception as exc:    # a bad frame must never stop the control loop
            self.get_logger().warn(f'frame skipped: {exc}', throttle_duration_sec=2.0)

    # ---- control loop -------------------------------------------------------------
    def tick(self):
        if self.finished:
            return
        if self.grab is not None and self.grab.latest() is not None:
            self.process_frame(self.grab.latest())
        now = self.now()
        dt = max(now - self.prev_tick, 1e-3)
        self.prev_tick = now
        visible = (now - self.det_time) <= self.detect_fresh
        obs = Observation(now=now, visible=visible, bearing=self.bearing if visible else 0.0,
                          distance=self.distance if visible else float('inf'), tof=self.tof)
        v, w = self.fsm.update(obs)
        self.v_cmd, self.w_cmd = v, w
        tw = Twist()
        tw.linear.x, tw.angular.z = float(v), float(w)
        self.cmd_pub.publish(tw)
        self.omegas = wheel_rates(v, w, self.r, self.lc)
        self.volts = self.voltage.step(self.omegas, dt)

        t_rel = now - self.t_start
        self._fast.writerow([f'{t_rel:.3f}', f'{v:.4f}', f'{w:.4f}', *[f'{x:.4f}' for x in self.volts], self.fsm.state])
        if self.fsm.state == ARRIVED:
            self.stop_ticks += 1
            if self.stop_ticks == 3:      # let the derivative term settle: voltages are exactly 0
                self.log_row(t_rel, f'ARRIVED ({self.fsm.arrival_reason})', visible)
                self.get_logger().info(f'Halted at {t_rel:.1f} s ({self.fsm.arrival_reason}); '
                                       f'log saved to {self.csv_path}')
                self.finished = True
                self._csv_file.close()
                self._fast_file.close()
            return
        if t_rel >= self.next_log:
            self.log_row(math.floor(t_rel + 1e-6), self.fsm.state, visible)
            self.next_log += 1.0

    def log_row(self, t, state, visible):
```

```
        row = [f'{t:.1f}' if isinstance(t, float) else t, *[f'{x:.3f}' for x in self.volts], state,
               f'{self.v_cmd:.3f}', f'{self.w_cmd:.3f}',
               f'{self.distance:.3f}' if visible else '', f'{self.bearing:.3f}' if visible else '',
               f'{self.tof:.3f}' if self.tof is not None else '',
               *[f'{x:.3f}' for x in self.omegas]]
        self._csv.writerow(row)
        self._csv_file.flush()
        self.get_logger().info(f'{row[0]:>5} s | V1..V4 = ' + ' '.join(f'{x:7.3f}' for x in self.volts)
                               + f' V | {state}')

    def shutdown(self):
        """Always leave the robot stopped."""
        for _ in range(3):
            self.cmd_pub.publish(Twist())
        if self.grab is not None:
            self.grab.running = False
        for f in (self._csv_file, self._fast_file):
            if not f.closed:
                f.close()


def main(args=None):
    rclpy.init(args=args)
    node = FlatnessNav()
    try:
        # exit by itself shortly after the robot has halted, so the caller does not wait for a timeout
        while rclpy.ok() and not node.finished:
            rclpy.spin_once(node, timeout_sec=0.1)
        t_end = time.time() + 1.0
        while rclpy.ok() and time.time() < t_end:
            rclpy.spin_once(node, timeout_sec=0.1)     # let the final zero commands go out
    except KeyboardInterrupt:
        pass
    finally:
        node.shutdown()
        node.destroy_node()
        if rclpy.ok():
            rclpy.shutdown()


if __name__ == '__main__':
    main()
```

## Build, test and run

```
source /opt/ros/jazzy/setup.bash
colcon build --symlink-install && source install/setup.bash
python -m pytest -q src/rm_flatness_nav/test                              # 18 tests
ros2 launch robomaster_ros ep.launch conn_type:=rndis                     # or conn_type:=ap
ros2 launch rm_flatness_nav flatness_nav.launch.py tag_id:=3 tag_size:=0.15 \
    csv_path:=$PWD/motor_voltage_log.csv
```